

Synergistic Inhibition of Mild Steel Corrosion in Hydrochloric Acid by Kecapi (*Sandoricum koetjape*) Leaf Extract and Iodide Ions

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Abstract

The corrosion inhibition performance of steel in hydrochloric acid solution using kecapi (*Sandoricum koetjape* (Burm.f.) Merr) leaf extract (KLE) and the synergistic effect of iodide ions was investigated using the weight loss method at temperatures ranging from 30 to 60 °C. The inhibition efficiency increased with increasing concentrations of both KLE and iodide ions, as well as with increasing temperature. A maximum inhibition efficiency of 86.61% was achieved at a KLE concentration of 6 g L⁻¹ in the absence of iodide ions, which further increased to 92.63% upon the addition of 0.5 g L⁻¹ iodide ions, indicating a pronounced synergistic effect. The adsorption behavior of KLE on the mild steel surface followed the Langmuir adsorption isotherm, suggesting monolayer adsorption. FTIR analysis revealed shifts in characteristic functional group bands, confirming interactions between extract constituents and the steel surface. SEM observations demonstrated significant improvement in surface morphology for mild steel specimens immersed in 1 N HCl containing KLE and iodide ions compared to the uninhibited solution. The synergistic effect between KLE and iodide ions was quantitatively evaluated using synergism parameters derived from inhibition efficiency values, confirming enhanced adsorption and corrosion protection.

Keywords

Sandoricum koetjape
mild steel
green inhibitor
iodide synergistic
Langmuir isotherm

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1. Introduction

Mild steel is extensively used in various industrial applications, including crude oil refining and petrochemical processes, owing to its favorable mechanical properties and cost effectiveness. However, steel is highly susceptible to corrosion, particularly in acidic environments such as hydrochloric acid solutions commonly employed during pickling, descaling, and acid cleaning operations. Direct contact between steel surfaces and corrosive media accelerates electrochemical reactions, leading to material degradation, economic losses, and potential safety hazards^[1].

The inherent vulnerability of mild steel to corrosion represents a major limitation to its long-term industrial application. Therefore, effective corrosion control strategies are essential to ensure sustainable utilization of steel in industrial and community infrastructures^[2]. Among the various protection methods, the use of corrosion inhibitors remains one of the most practical and economical approaches, especially for acidic environments.

Efficient corrosion inhibitors generally possess heteroatoms such as nitrogen, oxygen, sulfur, or phosphorus, along with conjugated π -electron systems that facilitate adsorption onto the metal surface through donor-acceptor interactions with vacant d-orbitals of iron atoms. These interactions promote the formation of a compact, adherent, and protective film that isolates the metal surface from the corrosive medium. In addition, modern corrosion inhibitors are expected to be cost-effective, biodegradable, and environmentally benign^{[3], [4]}. Consequently, increasing attention has been devoted to organic inhibitors derived from natural plant sources^[5].

Several studies have demonstrated the effectiveness of plant-based extracts as green corrosion inhibitors for steel in acidic media. For instance, *Gnetum gnemon* L. bark extract exhibited significant inhibition performance for steel in hydrochloric acid solution^[6], while rubber leaf extract achieved an inhibition efficiency of up to 86% for steel corrosion in HCl medium^[7]. These findings highlight the potential of phytochemical-rich plant extracts as

sustainable alternatives to conventional synthetic inhibitors.

In this context, kecap leaves were selected as a natural corrosion inhibitor. The extract of kecap leaves is rich in bioactive phytochemicals, including alkaloids, phenolic compounds, flavonoids, and triterpenoids, which exhibit strong adsorption affinity toward metal surfaces. Previous phytochemical investigations have identified san-drapsins A – E as major constituents of kecap leaves^[8], suggesting their potential contribution to corrosion inhibition through surface adsorption mechanisms.

The inhibition performance of plant extracts can be further enhanced through synergistic interactions with halide ions. Such synergism not only improves corrosion protection but also reduces the amount of organic inhibitor required^[9]. Among halide ions, iodide ions are particularly effective because of their large ionic radius, high polarizability, and strong adsorption affinity toward positively charged metal surfaces in acidic media. Adsorbed iodide ions can facilitate the adsorption of protonated organic molecules from plant extracts, leading to the formation of a more compact and stable protective layer. For instance, *Cichorium intybus* extract exhibited an inhibition efficiency of 74% for mild steel in 1 M HCl, which increased to 99% after the addition of iodide ions, demonstrating a strong synergistic effect between the extract constituents and iodide ions^[10].

To the best of our knowledge, no previous study has comprehensively investigated the synergistic corrosion inhibition performance of *Sandoricum koetjape* leaf extract in combination with iodide ions for mild steel in hydrochloric acid solution. Therefore, the objectives of this study were to evaluate the corrosion inhibition efficiency of kecap leaf extract (KLE), investigate the synergistic effect of iodide ions, and elucidate the adsorption and inhibition mechanisms involved. The inhibition performance was assessed through weight-loss measurements, adsorption studies, thermodynamic analysis, and surface characterization techniques. The findings are expected to provide new insights into the development of sustainable and highly efficient green corrosion inhibitor systems for acidic environments.

2. Material and Method

2.1. Materials

The materials used in this study were distilled water, hydrogen chloride (Smart-lab), mild steel, methanol, and *Sandoricum koetjape* (*Burm.f.*) Merr.

2.2. Instruments

The instruments used in this study were glassware, analytical balance (Ohaus C 214), magnetic stirrer, hot plate (AGIMATIC REV-E). The characterization instruments used were SEM (HitachiS-3400N), and FTIR (Perkin Elmer).

2.3. Methods

2.3.1. Material Preparation

Mild steel specimens ($2.0 \times 1.0 \times 0.1$ cm) with composition (wt%): 0.189 C, 0.141 Mn, 0.0715 Si, 0.0149 P, 0.019 Se, 0.0018 Cr, 0.0001 Ni, balance Fe, were used. The specimens were polished with SiC papers, rinsed with distilled water, degreased in acetone, dried at 60 °C, and weighed to obtain the initial mass (m_1).

2.3.2. Preparation of Extract (KLE)

Leaves of *Sandoricum koetjape* were washed, air-dried, and ground. About 500 g powder was macerated in 3000 mL methanol for 72 h. The filtrate was evaporated under reduced pressure to obtain the crude extract, which was diluted to desired concentrations.

2.3.3. Weight Loss Measurement

Specimens were immersed in 50 mL of 1 N HCl solution in the absence and presence of KLE (with and without iodide ions) at temperatures ranging from 30 to 60 °C for 7 h. After immersion, the samples were retrieved, rinsed, dried, and reweighed to obtain the final mass (m_2). The weight loss (Δm), corrosion rate, and inhibition efficiency were calculated using the following equations^[11].

Corrosion rate:

$$CR = \frac{m_1 - m_2}{A \times t} \quad (1)$$

where CR is the corrosion rate ($\text{mgcm}^{-2}\text{h}^{-1}$), m_1 and m_2 are the initial and final masses of the steel specimens (mg), respectively, A is the exposed surface area (cm^2), and t is the immersion time (hours).

Inhibition efficiency:

$$IE = \frac{CR_1 - CR_2}{CR_1} \times 100\% \quad (2)$$

where CR_1 and CR_2 represent the corrosion rates in the absence and presence of the inhibitor, respectively.

2.3.4. FTIR Analysis

Mild steel specimens were immersed for 7 days in the blank solution, KLE solution, and KLE–iodide system. The corrosion products and adsorbed films formed on the steel surface were subsequently collected by scraping,

mixed with KBr, and analyzed using FTIR spectroscopy. In addition, pure KLE was characterized under the same conditions to facilitate spectral comparison.

2.3.5. SEM Analysis

The surface morphology of mild steel specimens before and after 7 days of immersion was characterized using SEM. The unexposed specimen served as the reference, while the immersed specimens were used to evaluate the effect of the inhibitor on surface deterioration and film formation.

3. Results and Discussion

3.1. Weight Loss Measurement

Table 1 provides information on the effect of corrosion rate with variations in extract concentration and iodide concentration at temperatures of 30 - 60°C by weight loss method. In the Table 1, it can see that the value of the corrosion rate decreases with the increasing concentration of KLE. This proves that the increase in the concentration of the extract is caused by the presence of compounds contained in the extract which are adsorbed on the steel surface so that it slows down the corrosion rate^[12].

Table 1. Corrosion rate (CR) and inhibition efficiency (IE) of mild steel in 1 N HCl in the presence of different concentrations of KLE and KI at various temperatures

Concentration (g L ⁻¹)		Corrosion rate (mg cm ⁻² h ⁻¹) and inhibition efficiency (%)							
		30°C		40°C		50°C		60°C	
KLE	KI	CR	%IE	CR	%IE	CR	%IE	CR	%IE
Blank (HCl 1 M)	-	0.7699	0	1.4129	0	3.2441	0	4.7751	0
1	-	0.2317	69.91	0.3704	73.78	0.6832	78.94	0.7263	84.79
2	-	0.2307	70.04	0.3683	73.94	0.6372	80.36	0.7244	84.83
4	-	0.2205	71.36	0.3392	75.99	0.5571	82.83	0.7004	85.33
6	-	0.2038	73.54	0.3001	78.76	0.4947	84.75	0.6393	86.61
6	0.1	0.2186	71.28	0.2613	74.21	0.3692	80.66	0.4178	91.14
6	0.2	0.2168	71.51	0.2452	75.80	0.3663	80.81	0.4059	91.39
6	0.3	0.2103	72.37	0.2240	75.92	0.3530	81.50	0.3976	91.57
6	0.4	0.2061	72.92	0.2112	79.16	0.3509	81.61	0.3709	92.13
6	0.5	0.2024	73.41	0.2051	79.76	0.3361	82.39	0.3476	92.63

Conversely, the corrosion rate increases with increasing temperature due to the acceleration of electrochemical reactions. However, the inhibition efficiency also increases with temperature, reaching a maximum value of 86.61% at 6 g L⁻¹ KLE and 60 °C. This suggests that the adsorption process is thermally activated and involves strong inhibitor-metal interactions^[13].

The addition of iodide ions significantly enhances the inhibition efficiency, achieving 92.63% at 0.5 g L⁻¹ iodide ions. This improvement indicates a strong synergistic interaction between KLE and iodide ions, leading to increased surface coverage and improved corrosion protection. The increased inhibition efficiency in the presence of iodide ions is due to the cooperative reaction

between the inhibitor molecules and the KI. There is a competition on the adsorption on the active sites of the metal surface between KLE and I⁻ ions^[14].

3.2. Adsorption Consideration

The adsorption behavior of iodide ions on the mild steel surface was analyzed using the Langmuir adsorption isotherm^[15]:

$$C/\theta = 1/K_{\text{ads}} + C. \quad (3)$$

where C is the concentration of iodide ions, θ is the degree of surface cover ($\theta = \text{IE}/100$), and K_{ads} is the adsorption equilibrium constant. A linear relationship between C/θ and C (Figure 1) with a correlation coefficient ($R^2 \approx 0.9999$) confirms that the adsorption process follows the Langmuir model. This indicates monolayer adsorption on a homogeneous surface with no significant interaction between adsorbed species^{[16][17]}.

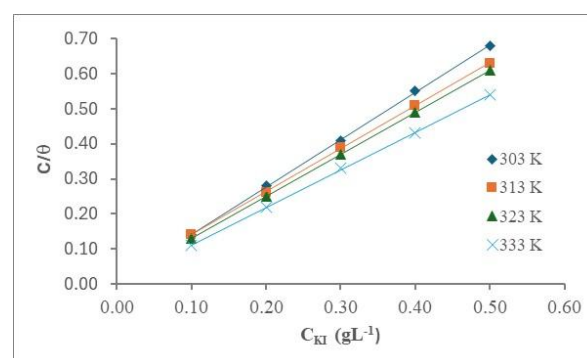


Figure 2. Langmuir adsorption isotherm plots (C/θ versus C_{KI}) for mild steel in 1 N HCl containing 6 g L⁻¹ KLE at different temperatures (303 – 333 K)

From Table 2 it can be seen that the value of K_{ads} increases with increasing temperature. This can be proven by increasing the percentage of the highest inhibition efficiency obtained at the highest temperature. The standard free energy of adsorption ΔG_{ads} was calculated using the following equation^[18]:

$$\Delta G_{\text{ads}} = -RT \ln(C_{\text{H}_2\text{O}} K_{\text{ads}}) \quad (4)$$

where R is the molar gas constant, T is the absolute temperature and $C_{\text{H}_2\text{O}}$ is the concentration of water in solution in 1000 g L⁻¹. The negative values of Gibbs free energy of adsorption (ΔG_{ads}) at all temperatures confirm that the adsorption of KLE – KI on the steel surface occurs spontaneously. The ΔG_{ads} values become more negative with increasing temperature, from -31.53 kJ mol⁻¹ at 30°C to -36.92 kJ mol⁻¹ at 60 °C. This trend indicates stronger and more stable adsorption at higher temperatures. The magnitude of ΔG_{ads} , which lies

between -20 and -40 kJ mol^{-1} , suggests that the inhibition mechanism involves mixed adsorption, combining physical adsorption through electrostatic attraction and chemical adsorption through donor-acceptor interactions between KLE molecules and the steel surface.

Table 2. Thermodynamic parameters for the adsorption of extracts on steel in 1 N HCl with the addition of iodide at different temperatures

Temperature (°C)	K_{ads}	ΔG_{ads} (kJmol ⁻¹)	ΔH_{ads} (kJmol ⁻¹)	ΔS_{ads} (Jmol ⁻¹)
30	73.2871	-31.53		
40	120.0215	-30.43	30.80	144.59
50	438.6940	-34.89		
60	619.0729	-36.92		

From the values of K_{ads} and ΔG_{ads} listed in Table 2, the standard enthalpy (ΔH_{ads}) and entropy (ΔS_{ads}) of adsorption were determined using the following thermodynamic relationship:

$$\ln K_{\text{ads}} = -\frac{\Delta H_{\text{ads}}}{R} \left(\frac{1}{T} \right) + \frac{\Delta S_{\text{ads}}}{R} \quad (5)$$

The values of ΔH_{ads} and ΔS_{ads} were obtained from the slope and intercept of the linear plot of $\ln K_{\text{ads}}$ versus $1/T$, and the results are summarized in Table 2.

The positive value of ΔH_{ads} indicates that the adsorption of inhibitor molecules onto the metal surface is an endothermic process, suggesting that adsorption is favored at higher temperatures. The positive value of ΔS_{ads} implies an increase in the degree of disorder during the adsorption process. This behavior is commonly associated with the substitution of pre-adsorbed water molecules by inhibitor molecules at the metal/solution interface. The desorption of water molecules into the bulk solution increases the overall randomness of the system, resulting in a positive entropy change. These findings support the spontaneous formation of an adsorbed protective layer on the metal surface through a displacement adsorption mechanism^{[19][20][21]}.

3.3. Kinetics and Thermodynamic Studies

Activation energy represents the minimum energy barrier that must be overcome for a corrosion reaction to proceed. It was evaluated using the Arrhenius equation^[22].

$$\ln CR = \ln A - \frac{E_a}{RT} \quad (6)$$

where E_a is the activation energy (kJ mol^{-1}), A is the Arrhenius pre-exponential factor, R is the universal gas constant, and T is the absolute temperature. Using equation (4) E_a was determined from the slope of the plot of

$\ln CR$ versus $1/T$. The linearity of this plot (Figure 3) confirms that the corrosion process follows Arrhenius behavior.

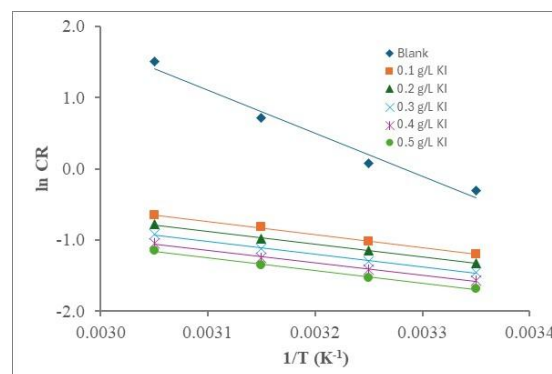


Figure 3. Arrhenius plot of $\ln CR$ versus $1/T$ (K^{-1}) for steel corrosion in 1 N HCl in the absence and presence of iodide ions added to 6 g L^{-1} KLE

Table 3. Activation energy (E_a), enthalpy of activation (ΔH^*) and entropy of activation (ΔS^*) values for mild steel corrosion in the absence and presence of iodide ions at an extract concentration of 6 g L^{-1}

Medium	E_a (kJmol ⁻¹)	ΔH^* (kJmol ⁻¹)	ΔS^* (Jmol ⁻¹)
Blank (HCl 1 N)	52.98	50.34	-81.06
HCl 1 N + 1 g/L extract	34.08	31.44	-153.00
HCl 1 N + 2 g/L extract	33.55	30.91	-154.84
HCl 1 N + 4 g/L extract	33.33	30.69	-156.14
HCl 1 N + 6 g/L extract	33.01	30.37	-158.00
HCl 1 N + 6 g/L extract + 0.1 g/L KI	19.22	16.58	-203.07
HCl 1 N + 6 g/L extract + 0.2 g/L KI	19.15	16.50	-203.53
HCl 1 N + 6 g/L extract + 0.3 g/L KI	19.13	16.49	-203.77
HCl 1 N + 6 g/L extract + 0.4 g/L KI	19.01	16.37	-204.63
HCl 1 N + 6 g/L extract + 0.5 g/L KI	17.73	15.09	-209.00

As presented in Table 3, the activation energy (E_a) decreases in the presence of KLE and further decreases upon the addition of iodide ions in 1 N HCl solution. This trend suggests that the inhibitor system modifies the corrosion pathway, facilitating the formation of an adsorbed protective layer on the steel surface. The reduction in E_a is commonly associated with chemisorption or strong interaction between inhibitor molecules and the metal surface, which alters the energy profile of the corrosion process. The positive values of the enthalpy of activation (ΔH^*) indicate that the dissolution of steel is an endothermic process. Moreover, the ΔH^* value for the blank solution is higher than that in the presence of KLE and iodide, implying that the energy barrier for the corrosion reaction is reduced when the inhibitor is present. This behavior supports the hypothesis that adsorption of inhibitor molecules facilitates the formation of a more stable activated complex. The entropy of activation (ΔS^*) provides insight into the degree of disorder during the formation of the activated complex. The negative values of

ΔS^* observed in this study indicate a decrease in randomness as the system transitions from reactants to the activated state. This suggests that the activated complex [23][24].

3.4. Synergistic Effect

The synergistic effect represents an effective strategy to enhance corrosion inhibition performance in aggressive media while simultaneously reducing the required dosage of organic inhibitors. In acidic solutions, the addition of iodide ions has been widely reported to significantly improve the inhibition efficiency of organic inhibitors through cooperative adsorption mechanisms^{[9], [10]}.

The existence and magnitude of the synergistic interaction between KLE and iodide ions were quantitatively evaluated using the synergism parameter (S), calculated according to the following expression^[25]:

$$S = \frac{1-\theta_{1+2}}{1-\theta'_{1+2}} \quad (7)$$

where

$$\theta_{1+2} = (\theta_1 + \theta_2) - (\theta_1 \cdot \theta_2). \quad (8)$$

Here, θ_1 and θ_2 represent the surface coverage due to iodide ions and KLE, respectively, while θ'_{1+2} denotes the experimentally determined surface coverage of the combined inhibitor system. The calculated synergism parameters (S) at different temperatures and iodide ion concentrations are summarized in Table 4.

Table 4. The value of the synergistic effect of KLE with iodide ion

Iodide Concentration (g L ⁻¹)	Synergistic Effects (S)			
	30°C	40°C	50°C	60°C
0	-	-	-	-
0.1	1.31	1.33	1.41	2.36
0.2	1.17	1.19	1.21	1.89
0.3	1.14	1.15	1.03	1.57
0.4	1.13	1.14	1.01	1.22
0.5	1.04	1.09	1.03	1.07

As shown in Table 4, all S values are greater than unity ($S > 1$) across the investigated temperature range, confirming the presence of a pronounced synergistic effect between KLE and iodide ions. The highest synergism values were observed at lower iodide concentrations (0.1 - 0.2 g L⁻¹), particularly at elevated temperatures, indicating optimal cooperative interactions under these conditions.

The decrease in S values at higher iodide concentrations suggests that excessive iodide ions may partially saturate the steel surface, limiting the availability of active adsorption sites for organic inhibitor molecules. This behavior is consistent with previously reported

observations for plant extract-halide systems, where an optimum halide concentration maximizes synergistic efficiency^[10].

The synergistic effect can be explained by the strong adsorption affinity of iodide ions toward the positively charged steel surface in acidic media. Due to their large ionic radius and high polarizability, iodide ions are preferentially chemisorbed onto the metal surface, creating localized negative charge centers. These negatively charged sites subsequently promote the electrostatic attraction and adsorption of protonated organic species present in KLE, thereby increasing surface coverage and strengthening the protective film.

Furthermore, the observed enhancement in inhibition efficiency upon iodide addition is consistent with the thermodynamic results, where more negative ΔG_{ads} values were obtained for the KLE-iodide system. This indicates stronger and more stable adsorption, involving a mixed physisorption-chemisorption mechanism. Similar synergistic behavior has been reported for other plant-based inhibitors combined with iodide ions in acidic environments^{[9], [10]}.

Overall, the synergistic interaction between KLE and iodide ions significantly improves corrosion protection by facilitating co-adsorption, increasing surface coverage, and stabilizing the adsorbed inhibitor film. This synergism not only enhances inhibition efficiency but also supports the development of effective, low-dose, and environmentally friendly corrosion inhibitor systems.

3.5. FTIR Analysis

FTIR analysis was performed to identify the functional groups of organic compounds present in KLE that contribute to corrosion inhibition through adsorption onto the steel surface^[26]. The FTIR spectra of pure KLE were compared with those of the surface films formed on steel specimens immersed in 1 N HCl solution containing 6 g L⁻¹ KLE, both in the absence and presence of 0.5 g L⁻¹ iodide ions, as shown in Figure 4.

As evident from Figure 4, noticeable shifts in several characteristic absorption bands are observed after steel immersion in inhibitor-containing solutions. These shifts confirm the interaction between KLE constituents and the steel surface, leading to the formation of an adsorbed protective layer that effectively suppresses the corrosion process^[27].

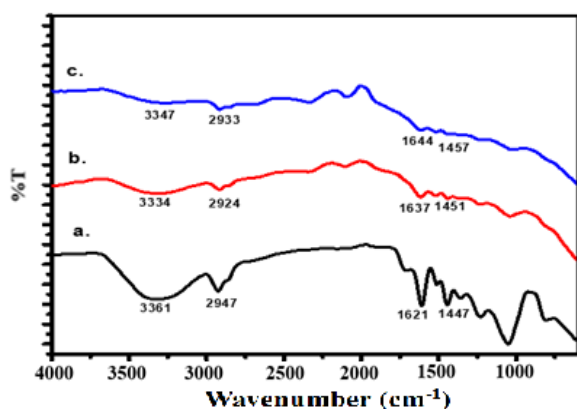


Figure 4. FTIR spectra of (a) KLE, (b) corrosion products formed on steel in 1 N HCl containing 6 g L⁻¹ KLE, and (c) corrosion products formed in 1 N HCl containing 6 g L⁻¹ KLE and 0.5 g L⁻¹ iodide ions

Specifically, the broad O–H stretching vibration observed at 3361 cm⁻¹ in the spectrum of pure KLE shifted to lower wavenumbers at 3334 cm⁻¹ and 3347 cm⁻¹ in the spectra of the surface films, indicating the involvement of hydroxyl groups in adsorption through hydrogen bonding or coordination interactions with the steel surface. The aliphatic C–H stretching vibration initially appearing at 2947 cm⁻¹ shifted to 2924 cm⁻¹ and 2933 cm⁻¹, suggesting changes in the molecular environment of alkyl groups upon adsorption.

In addition, the C=C stretching vibration associated with aromatic or conjugated systems shifted from 1621 cm⁻¹ to higher wavenumbers at 1637 cm⁻¹ and 1644 cm⁻¹, reflecting π -electron interactions between aromatic rings in KLE constituents and the vacant d-orbitals of iron atoms. Furthermore, the C–H bending vibration shifted from 1447 cm⁻¹ to 1451 cm⁻¹ and 1457 cm⁻¹, providing additional evidence of structural rearrangement following adsorption.

The observed band shifts, particularly in the O–H and C=C functional groups, indicate strong interactions between KLE molecules and the steel surface. These interactions are responsible for the formation of a compact and adherent protective film that isolates the metal surface from aggressive chloride ions. The presence of iodide ions further enhances these interactions by promoting co-adsorption and stabilizing the adsorbed inhibitor layer, which is consistent with the synergistic inhibition behavior observed in weight loss and thermodynamic analyses.

3.6. SEM Analysis

Scanning Electron Microscopy (SEM) was employed to examine the surface morphology and structural changes

of the mild steel samples. As presented in Figure 5(a), the untreated mild steel surface exhibits a relatively smooth and uniform appearance, with visible scratches attributed to the mechanical polishing process. The surface remains largely intact and flat prior to exposure to the corrosive medium.

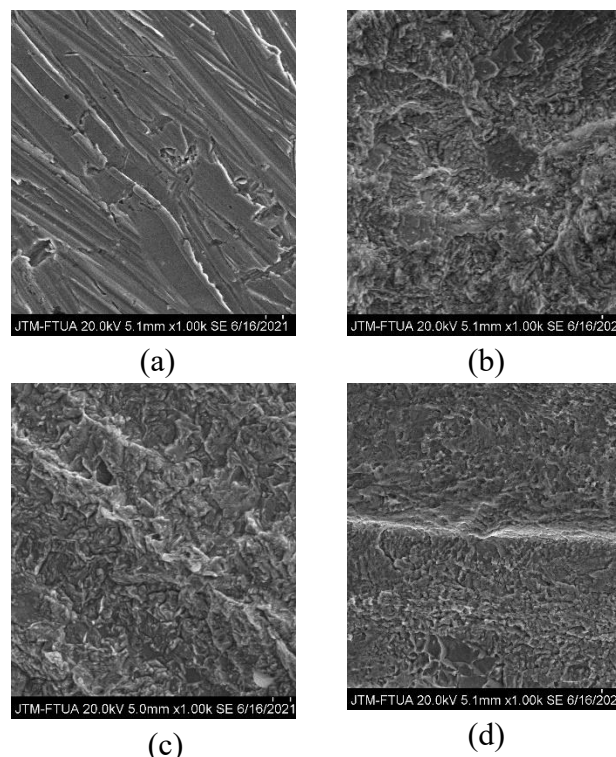


Figure 5. SEM micrographs of the steel surface: (a) untreated, (b) immersed in 1 N HCl, (c) immersed in 1 N HCl with 6 g L⁻¹ extract, and (d) immersed in 1 N HCl with 6 g L⁻¹ extract and 0.5 g L⁻¹ iodide ions for 7 days

In contrast, Figure 5(b) shows a severely deteriorated surface after immersion in the corrosive solution, characterized by the presence of pits, cavities, and irregular features. These defects indicate significant corrosion damage resulting from the aggressive interaction between the metal surface and the acidic environment.

The morphology of the mild steel surface in the presence of KLE, as illustrated in Figure 5(c), appears considerably improved, with fewer visible defects. This suggests that the extract molecules are adsorbed onto the steel surface, forming a protective layer that mitigates the corrosion process.

Furthermore, as shown in Figure 5(d), the addition of potassium iodide leads to a more uniform and compact surface with the absence of noticeable pits or cavities. This observation indicates enhanced surface protection, which can be attributed to the synergistic adsorption of KLE constituents and iodide ions, resulting in the

formation of a stable and adherent protective film that effectively inhibits corrosion^[28].

4. Conclusion

Kecapi leaf extract (KLE) was demonstrated to be an effective green corrosion inhibitor for steel in hydrochloric acid solution, with inhibition efficiency increasing alongside extract concentration due to enhanced surface adsorption. Despite the increase in corrosion rate with temperature, the inhibition efficiency also improved, indicating stronger inhibitor–metal interactions at elevated temperatures. The maximum inhibition efficiency reached 86.61% at 6 g L⁻¹ KLE and 60 °C, which further increased to 92.63% upon the addition of 0.5 g L⁻¹ iodide ions, confirming a significant synergistic effect. Adsorption followed the Langmuir isotherm, supported by FTIR, and SEM analyses, which confirmed the formation of a protective and compact inhibitor film. Overall, the KLE – iodide system provides an efficient and environmentally friendly corrosion inhibition strategy with potential industrial applicability.

Author contributions

Conceptualization: E.E, S.S.

Data curation: S.W

Formal Analysis: E.E, S.W, S.S

Funding acquisition: E.E, S.W.

Investigation: S.W

Methodology: E.E, S.S.

Project administration: S.W

Resources: E.E

Software: E.E, S.W

Supervision: E.E, SW

Validation: E.E, S.W, S.S

Visualization: E.E, S.W.

Writing – original draft: S.W

Writing – review & editing: E.E, S.S

Conflict of interest

The authors declare no conflict of interest.

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